

DEBAYAN DAS

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[LinkedIn](#) • [GitHub](#) • [LeetCode](#)

EDUCATION

Jadavpur University – B.E., Electronics & Tele-Communication Engineering
CGPA: 8.2 / 10

2023 – 2027

TECHNICAL SKILLS

Languages: C/C++, Rust, Python, JavaScript (ES6+), HTML5, CSS3, SQL.
Frameworks: React.js, Node.js, Express.js, RESTful APIs, Responsive UI Design
Databases: MongoDB, MySQL, Neon Postgres.
Tools: Git, GitHub, VS Code, Linux
Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management System, Operating Systems

PROFESSIONAL EXPERIENCE

AlgoBengal Research Labs · *Cryptography Research Intern* ·
Kolkata, India (On-site)

June 2025 – Present

- Architected a pure-Rust port of the CRYSTALS-Dilithium post-quantum signature scheme from C, enforcing zero unsafe code to guarantee complete memory safety.
- Proved byte-for-byte functional equivalence against the C reference implementation by engineering and validating bi-directional FFI interoperability.

PROJECTS

SpeedAI – AI SaaS Platform [[Live ↗](#)][[Code ↗](#)] | *React, Node.js, Express, Neon Postgres*

- Developed a modern **AI SaaS web app** (**MERN-stack** equivalent using **Neon Postgres**) providing users with AI image generation, article writing, and a community platform.
- Integrated **Clerk** for secure authentication, **Cloudinary** for optimized media management, and built a highly responsive frontend using **React 19**, **Vite 8**, and **Tailwind CSS**.

Heuristic Search Based Rubik's Cube Solver | *C++, Bitmasking, IDA**

- Designed a high-performance 3x3 Rubik's Cube solver by modeling the cube as a Bitboard, utilizing advanced OOP principles and bitmasking to enable efficient state transitions.
- Implemented the IDA* (Iterative Deepening A*) algorithm, guaranteeing optimal solutions within 15 moves with an average solve time of under 10 seconds.
- Engineered a custom heuristic database using BFS and a permutation indexer to significantly accelerate search depth and performance.

High-Performance Cache Simulator [[Code ↗](#)] | *C++, Multi-threading, STL*

- Engineered a thread-safe concurrent cache engine in C++ implementing **LRU** and **LFU** eviction policies, using **std::mutex** and templates to ensure safe parallel read/write operations across multi-threaded execution.
- Built a synthetic workload generator modeling an **80/20 data locality pattern**; benchmarked over **200,000 parallel operations**, dynamically logging hit/miss ratios, eviction rates, and latency metrics to compare algorithm efficiency.

ACHIEVEMENTS

- Solved **500+ DSA problems** across LeetCode, Codeforces, and other competitive programming platforms.
- Achieved **Knight rating (Highest Rating: 1993)** on **LeetCode**, placing in the top competitive programming tier.
- Selected as a **Siemens Scholar** from over 300 candidates across India, recognizing academic and technical excellence.